Personalized Interest Graphs for Theme-Driven User Behavior

Oded Zinman
eBay Inc.
Netanya, Israel
ozinman@ebay.com

Nazmul Chowdhury
eBay Inc.
Nyack, New York, USA
nchowdhury@ebay.com

Leandro Fiaschetti
eBay Inc.
Tandil, Buenos Aires, Argentina
lfiaschetti@ebay.com

Yuri M. Brovman
eBay Inc.
Miami, Florida, USA
ybrovman@ebay.com

Guy Feigenblat
eBay Inc.
Netanya, Israel
gfeigenblat@ebay.com

Yotam Eshel
eBay Inc.
Netanya, Israel
yeshel@ebay.com

Abstract

Many eBay users turn to our platform to pursue theme-centric interests that span diverse product categories—for example, a Star Wars fan might search for related video games, toys, memorabilia, and artwork. Existing recommendation systems, typically optimized for short-term engagement, often fail to surface cross-category items aligned with these deeper interests. We present an end-to-end recommendation framework built around a user-interest graph generated by LLM chain. The graph captures user preferences at multiple levels of granularity, enabling a balance between relevance-driven and serendipity-driven recommendations. The system has been deployed at scale, serving millions of users across billions of items. An online A/B test on the eBay homepage showed a significant improvement in engagement with previously unseen categories, alongside gains in purchases and buyer count.

CCS Concepts

• **Computing methodologies → Natural language generation**.

Keywords

Recommendation, Large Language Models, Knowledge Graph

ACM Reference Format:
Oded Zinman, Nazmul Chowdhury, Leandro Fiaschetti, Yuri M. Brovman, Guy Feigenblat, and Yotam Eshel. 2025. Personalized Interest Graphs for Theme-Driven User Behavior. In *Proceedings of the Nineteenth ACM Conference on Recommender Systems (RecSys '25), September 22–26, 2025, Prague, Czech Republic.* ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/3705328.3748133

1 Introduction

Online marketplaces, such as eBay, often cater to large numbers of shoppers with diverse needs. While many visit with a directed intent—buying a specific item like a cellphone, picture frame, or chair—others have broader interests, such as fishing, DIY, or themed merchandise. These shoppers show less directed patterns and may browse products revolving around a central theme. For example, a Chicago Bulls fan might treasure hunt for autographed game shirts, vintage posters, and trading cards, while a DIY enthusiast might explore various power tools, hardware, and niche devices. Unlike transient shopping missions, theme-oriented interests tend to persist over time and can be effectively leveraged for personalization and inspiration-oriented recommendations.

Product-oriented interests—focused on a single category— are easier to detect. For instance, users who repeatedly visit the sneakers category can be identified as "sneakerheads." However, many interests are theme-oriented, spanning multiple, often loosely unrelated, categories. A fitness enthusiast, for example, might engage with gear, supplements, wearables, and books—all tied to one interest but spread across categories.

Current recommendation systems often favor staying within category boundaries, largely due to their optimization for short-term engagement signals like next-click prediction. This emphasis on immediate relevance can come at the expense of serendipity [2, 7]. The challenge is to surface items aligned with a user's core interests—even from unfamiliar categories—without sacrificing relevance or user experience.

To address this challenge, we propose a holistic LLM-chain-based recommendation system that spans the full pipeline from user interest modeling to item retrieval. Users are modeled as graphs, where nodes represent interests at varying levels of granularity. This structure enables dynamic traversal from broad themes to specific shopping missions, balancing relevance and serendipity. The system combines LLMs' world knowledge and summarization abilities with collaborative filtering signals to keep interests grounded in real user behavior.

A number of existing works have used LLMs to conceptualize user behavior patterns. Christakopoulou et al. [3] clustered YouTube videos into coherent user journeys and used LLMs for cluster labeling, but did not leverage these journeys for downstream recommendations. A related effort by Google DeepMind [8] fine-tuned an LLM on user data to predict engagement with new inventory clusters based on past interactions. However, their model relies on static, global interests, lacking personalization to user-specific preferences. Another line of work models user intentions through knowledge graphs [1, 5, 9]. FolkScope [9], constructs a large-scale intention graph using LLMs and human refinement. However, as noted by Zhou et al. [11], FolkScope is characterized by category-rigidity, making it less suitable for our use case.

Our paper presents a scalable, end-to-end system deployed at eBay, serving millions of users across billions of items. An A/B test

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RecSys '25, Prague, Czech Republic
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ACM ISBN 979-8-4007-1364-4/25/09
https://doi.org/10.1145/3705328.3748133

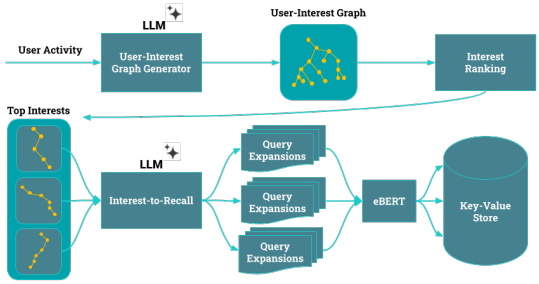

Figure 1: Production near-real-time LLM chain

within a homepage module showed a +17.02% lift in the inspiration metric (clicks on previously unseen categories), alongside uplifts in purchases and buyer count (+0.25% and +0.14%, respectively), indicating both novelty and maintained relevance.

2 Architecture

The high-level LLM chain architecture and its key building blocks are illustrated in Figure 1. This section presents a system-level overview, while the following sections delve into the details of each module. To mitigate LLM latency, the LLM chain is executed via a near-real-time pipeline. The system processes activity logs and stores embeddings for online item retrieval.

The pipeline is triggered upon processing new user activity. The user's current and historical site interactions are submitted to a Kafka queue, which activates the *User-Interest Graph Generator*—an LLM module that constructs the user's interest graph, as detailed in section 2.1. This graph is decomposed into linear paths, referred to as *interests* (e.g., "Sci-fi Movies → Memorabilia → Posters → Star Wars"). These interests are ranked by a machine learning model trained to predict future engagement, as described in section 2.2. The top-ranked interests are then passed to the *Interest-to-Recall* LLM (section 2.3), which generates query expansions aimed at retrieving a diverse set of related inventory items. Each query expansion is processed by eBERT—a BERT-based model [4] fine-tuned on eBay data—to map it into the inventory embedding space. Both the raw expansions and their embeddings are stored in a key-value store for online retrieval.

Candidate item recall and ranking occur in the online setting—e.g., when a user enters the homepage and triggers a request to populate the recommendation module. We retrieve the user's top string query expansions and their embeddings from a key-value store. The raw strings are used to fetch candidates via eBay's search engine, while the embeddings are used in an approximate nearest neighbor search powered by eBay's KNN service. The recalled items—typically in the hundreds—are then ranked by a deep learning model, narrowing the set to a few dozen high-quality recommendations.

2.1 User-Interest Graph

The graph's structure plays a central role: the top-level nodes represent broad, often cross-category interests, while the bottom-level nodes correspond to specific shopping missions. As such, traversing the graph enables multi-level exploration of user preferences. Figure 2 illustrates a toy example of such a user-interest graph. The graph is constructed from the user's site activity, specifically their search queries—concise expressions of intent. The graph is modeled as a directed acyclic graph (DAG). Higher-level nodes capture broad interests, while lower-level nodes represent increasingly specific interests. The sink nodes denote the narrowest interests and correspond to concrete search queries submitted by the user. Edges represent semantic relationships between interests, as inferred from the user's activity. For instance, as depicted in figure 2, the edge 'Sci-fi Movies → Memorabilia' refines the user's engagement with Sci-fi Movies, suggesting an affinity for collectibles related to the genre. Thus, a node's meaning is best understood in the context of its ancestor nodes.

Each node is associated with two properties: (a) Interest Name: An open-ended description of the interest, generated by the LLM. (b) Interest Types: Tags that categorize the interest (shown in square brackets in Figure 2). These tags are selected from a set of 35 categories, providing structure to the free-form interest names.

The user-interest graph generation task is both nuanced and token-intensive. Moreover, graphs must be continuously updated as users engage with the platform. While proprietary LLMs such as Gemini and GPT are effective in-context learners capable of generating high-quality graphs, their inference costs are prohibitive at scale. To balance quality with cost and scalability, we adopted a teacher–student paradigm. A proprietary LLM (teacher) generated user-interest graphs used to fine-tune a smaller, open-source student model.

For the teacher model, we used Gemini Flash 1.5 with an in-context prompt that followed a step-by-step process, generalizing from specific search queries to broader user interests before constructing the graph: (a) Write out 'thoughts' about the user's preferences; (b) Identify distinct shopping missions and their underlying passions; (c) Extract primary buyer passions across these missions. Only after completing this reasoning process does the LLM generate the user-interest graph, represented as an S-expression—a compact textual format well-suited for structured output.

For the student model, we trained a LoRA adapter [6] on top of Mistral 7B Chat, fine-tuned to generate S-expressions in the same format as the teacher. Fine-tuning the student model significantly reduced both input and output token counts. On the input side, prompt simplification and removal of few-shot examples caused a drastic 94% reduction of the average input tokens. On the output side, training the student to generate the S-expression directly (omitting the preceding reasoning) reduced the average output by 79%. The student model is deployed across 33 A100 GPUs and generates over 10 million user-interest graphs daily.

We built an interactive tool to explore individual graphs. Figure 3 illustrates how this tool enables zoom-in analysis of parts of the graph. The full graph is on the bottom-right corner.

2.2 Interest Ranking

At this stage, we incorporate user data and machine learning to rank user interests based on their likelihood of re-engagement, giving priority to cross-category interests. This approach leverages collaborative filtering to anchor the interests in actual user behavior.

The user-interest graph is first decomposed into distinct interests, where each interest corresponds to a linear path starting from the

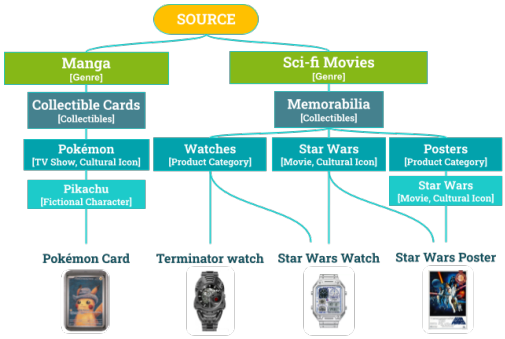

Figure 2: User-Interest graph representation

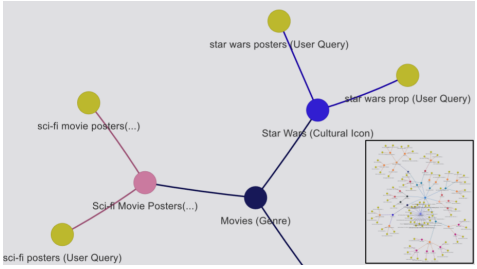

Figure 3: Interactive interface of a user's graph

source node. These paths can represent varying levels of granularity. For example, the graph in Figure 2 includes both a highly granular path— Sci-fi Movies→Memorabilia→Posters→Star Wars —and a more coarse-grained path— Sci-fi Movies→Memorabilia→Posters.

We trained the ranker to predict future user engagement. Using historical site interactions, we generated user-interest graphs and converted them into linear paths, as described above. Each site activity (sink node) was tagged with the path leading to it. Each user's interest-tagged activities were split to exclude their last session, which served as the prediction target. Given the user context and interest, the ranker was trained to predict whether the user would revisit the same interest in the final session. We used an XGBoost model with features such as the number of sink nodes in the subgraph rooted at the interest. To promote interests spanning category boundaries, we applied sample weighting—emphasizing cases where the predicted interest crossed into new categories. Evaluation with NDCG and MRR showed an 11% lift over baselines based on activity recency or frequency.

2.3 Interest-to-Recall

The top-ranked interests are passed to the *Interest-to-Recall* LLM. Each interest is provided separately as input to the LLM, which outputs a set of k query expansions (k=*10*) designed to include cross-category products aligned with the e-commerce inventory. It leverages the LLM's world knowledge to translate interest representations into actionable search queries. For example, feeding Sci-fi Movies→Memorabilia→Star Wars results in Star Wars Funko Pops, Star Wars Lego Sets, Star Wars The Black Series Action Figure, etc.

Model	Edge	Node	Task Alignment	Safety
Teacher	**6.61**	**33.70**	24.21	0.02
Student	7.33	34.01	**23.23**	0.02

Table 1: Errors identified by the LLM-as-a-judge

The number of calls to Interest-to-Recall converges over time, new paths become increasingly rare. Interest-to-Recall requires relatively short and simple prompts. Therefore, we opted to use the proprietary Gemini Flash 1.5 model directly via in-context learning, without applying the teacher–student paradigm. As described in section 2, the output query expansions are fed into eBERT to produce embeddings used online to retrieve live items recall.

3 Online A/B Test

We evaluated the framework via a two-week A/B test on the homepage module titled Inspired by Your Interests. The only difference between control and treatment groups was this module: (a) Control – based on eBay's existing cross-category algorithm using structured data; (b) Treatment – powered by our proposed method. Results showed a +17.02% lift in the inspiration metric—clicks on items from previously unexplored categories—indicating better exposure to novel product categories. The treatment also drove gains in purchases (+0.25%) and buyer count (+0.14%), highlighting that our approach effectively addresses theme-centric, cross-category interests without compromising relevance. While novelty and relevance often trade off, our method may boost both by reducing homepage recommendation homogeneity.

4 Offline Evaluation

We conducted an offline evaluation focused on assessing the user-interest graph, a core module in our framework. Using the *LLM-as-a-judge* approach [10], we measured the quality and safety of the generated graphs. To facilitate comprehensive evaluation—especially given that each graph can contain dozens of nodes—we decomposed them into smaller, manageable subgraphs before presenting them to the LLM judge (GPT-4). The LLM was tasked with assessing graph quality, detecting hallucinations (e.g., unrelated edges or misclassified interest types), flagging unsafe recommendations, and evaluating task alignment (e.g., determining if the interests reflect deep passions rather than generic behavior).

To validate this process, we asked domain experts to answer the same evaluation questions. Their responses showed strong correlation with the LLM's, reinforcing the reliability of the automated judging method. Table 1 shows the LLM's assessment of both teacher and student models, revealing only minor performance degradation in the student model.

5 Conclusion

We presented a scalable, LLM-driven recommendation system that models users' theme-oriented interests as graphs. Combining multi-level user modeling, collaborative filtering, and LLM-based query generation, the system balances relevance and serendipity to support cross-category exploration. Online A/B tests show improved engagement, discovery of novel content, and increased commercial value.

6 Speaker Bio

Oded Zinman is an Applied Researcher at eBay, where he has been a member of the Buyer Experience AI team since 2020. His work primarily focuses on NLP-based recommendation systems. He holds both a Master's and a Bachelor's degree from Ben-Gurion University of the Negev, with an academic focus on applied statistics and machine learning.

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